MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL
DEPARTMENT OF ELECRICAL ENGINEERING
END-TERM EXAMINATION, NOV.-2025

Course: B. Tech. **Branch: Electrical Engineering**

Semester: III,

Subject: EM Fields & Materials **Subject code:** EE-24211

Time: 2:00 Hours **Max. Marks:** 40

NB: Attempt all Questions. Assume missing data if necessary. Both parts of a question should be solved at one place only.

I(a)	Derive the generalized relation for energy stored in a electric fields of continuous charge distribution.	05	CO2
I(b)	Solve the potential function between the conductors of coaxial cable by Laplace's equation, if V_0 is the potential difference between them. Also determine the expression for the electric field intensity at any point in the system.	05	CO2
II(a)	Verify that the field $\vec{A} = yz\,\hat{x} + zx\,\hat{y} + xy\,\hat{z}$ is both irrotational and solenoidal.	05	CO1
II(b)	Two extensive homogeneous isotropic dielectrics meet on plane $z=0$, for $z \geq 0$, $\varepsilon_{r1} = 4$, and for $z \leq 0$, $\varepsilon_{r2} = 3$. A uniform electric field $E_1 = 5x - 2y + 3z$ kV/m exist for $z \geq 0$. Determine Electric field for $z \leq 0$	05	CO2
III.(a)	State and prove the Maxwells equation based on Ampere's law. Also derive the modified form of this Maxwells equation.	05	CO3
III.(b)	For a uniform plane wave determine the ratio of magnitude of Electric field to magnetic field in air medium.	05	CO3
IV(a)	Discuss properties and application of soft and hard magnetic materials.	05	CO1
IV(b)	List the important resistor materials and discuss their properties and applications.	05	CO1

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